scoup.ai

Venture pipeline: calibrated ranking and two build plans

2026-10-18

> **Pre-product. Synthetic data only. Not investment advice.**

Documents in this volume:

- _scoring/FINAL.md
- _builds/commission-reconciliation/plan.md
- _builds/premium-audit-defense/plan.md

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Final calibrated venture ranking (venture-calibrator, 2026-10-18)

Scope: the 13 leading candidates from batch-1/2/3, re-scored by one hand on the seven RUBRIC.md criteria. Inputs: RUBRIC.md, batch-1.md, batch-2.md, batch-3.md, the 13 cards, freight-recovery/ARCHITECTURE.md (for platform reuse), and live checks on 2026-10-18.

Method limit: the session-wide WebSearch budget (200 calls, shared with the batch scorers) ran out after four searches. The rest was checked with WebFetch on the cited URLs and on primary documents (Federal Register API, a CBP court declaration PDF, 42 USC 300gg-115). Pages that returned 403 or did not load count as NOT FOUND, and under the rules an unverifiable claim lowers the score.

1. Verification table

Status: VERIFIED = seen on the page, or in the primary document where one is named. VERIFIED (secondary) = a law firm or trade press quoting the primary document. NOT FOUND = not on the cited page, or the page could not be read. CONTRADICTED = the source says something different.

#	Venture	Claim on card	Status	Source line
1	customs	US Supreme Court held on 2026-02-20 that IEEPA does not authorize tariffs	VERIFIED (secondary)	6-3, Learning Resources v. Trump, Roberts C.J. Source: Thomson Reuters, Supreme Court rejects IEEPA as a tariff statute, https://www.thomsonreuters.com/en-us/help/onesource-global-trade/regulatory-insights/2026/february-20th/supreme-court-rejects-ieepa-as-a-tariff-statute, accessed 2026-10-18 (search result); McCarter & English, https://www.mccarter.com/insights/ieepa-tariffs-held-unlawful-section-122-tariffs-commence-trade-deals-likely-to-falter/, accessed 2026-10-18 (search result). Opinion text not fetched.
2	customs	~330,000 importers, ~$166B, ~53M entries (CBP declaration, 2026-03-06)	VERIFIED (secondary)	"More than 330,000 importers have paid IEEPA duties", "Over 53 million entries include IEEPA duties", "CBP has collected approximately $166 billion in IEEPA duties and duty deposits"; declaration of Brandon Lord, 2026-03-06. Source: Dickinson Wright, CBP Proposes IEEPA Duty Refund Process, https://www.dwlaw.com/news-alerts/client-alert-ieepa-duty-refund-process, accessed 2026-10-18
3	customs	CAPE refund process; Phase 1 opened 2026-04-20	VERIFIED (primary)	"The first phase of the CAPE functionality became available ... at 7:11am eastern time on Monday, April 20, 2026." Source: CBP (Brandon Lord), Declaration responding to July 2, 2026 court order, CIT No. 25-00595, ECF 45, filed 2026-07-13, https://storage.courtlistener.com/recap/gov.uscourts.cit.17610/gov.uscourts.cit.17610.45.0.pdf, accessed 2026-10-18
4	customs	~2.1M entries failed validation	VERIFIED (secondary) for 2026-04-26; later primary figure is larger	2,124,394 entries failed entry validation as of 2026-04-26 (Lord declaration of 2026-04-28). Source: BDO / CBH search results, https://www.cbh.com/insights/alerts/ieepa-tariff-refund-update-cbp-files-update-on-cape-activity-in-euro-notions/, accessed 2026-10-18. Primary update: 4.77M entries failed by 2026-07-10, mainly because the entry was past CBP's 90-day reliquidation authority, had no Chapter 99 HTS number, or had already been filed. Source: CBP declaration ECF 45 (row 3)
5	customs	"11.2 million entries claimed in under a week"	CONTRADICTED (wording)	The 11,222,927 figure counts entries that passed entry validation and were accepted, not entries claimed. Source: CBH (row 4), accessed 2026-10-18 (search result)
6	customs	Card frames the $166B as an open refund pool	OUT OF DATE	By 2026-09-17, 286,000+ CAPE declarations had been filed, $134.7B accepted and ~$122B (incl. interest) certified to Treasury; 20,184 refunds ($1.3B) were on hold for missing ACH data; Phase 3 (finally liquidated entries under court-ordered reliquidation) started 2026-10-06. Source: C.H. Robinson, CBP Reports Major Progress on IEEPA Refunds; CAPE Phase 3 Launches October 6, https://www.chrobinson.com/es-us/resources/insights-and-advisories/client-advisories/2026q3/09-17-26-cbp-cape-phase-3-ieepa-refunds/, accessed 2026-10-18

#	Venture	Claim on card	Status	Source line
7	customs	Alliance: 2% of refund, $10k min, $100k cap	VERIFIED	"2% of your IEEPA refund (before interest), subject to a $10,000 minimum and a $100,000 cap." Source: Alliance CHB, IEEPA refund calculator, https://alliancechb.com/ieepa-refund-calculator/, accessed 2026-10-18
8	customs	Drawback contingency 15-30%; a second passage gives 10-25%	VERIFIED (15-30%); NOT FOUND (10-25%)	"Most providers charge a contingency fee of 15-30% of recovered duties." Source: Ginger Control (vendor blog), https://gingercontrol.com/blog/duty-drawback-service-how-to-choose-provider, accessed 2026-10-18
9	provider-data	CMS-4208-F2 requires MA directory submission to CMS "by 2027"	CONTRADICTED (date)	Rule text: "For plan years beginning on or after January 1, 2026, MA organizations must ... (3) Update the information ... within 30 days ... (4) Attest at least annually." 42 CFR 422.111(m); effective 2025-11-17; applicability 2026-01-01. CMS estimates a one-time cost of ~$797 and an ongoing cost of ~$253 per year per entity (700 entities). Source: CMS, Federal Register 90 FR 45140, doc 2025-18236, docket CMS-4208-F2, https://www.federalregister.gov/documents/full_text/text/2025/09/19/2025-18236.txt, accessed 2026-10-18
10	provider-data	CMS review found 48.74% of MA directory locations inaccurate	NOT FOUND	Not in the rule text. The cited vendor page (Assured) gives "40% to 60%" instead. Source: Assured, https://www.withassured.com/blog/the-no-surprises-act, accessed 2026-10-18. Ideon blog not re-fetched.
11	provider-data	No Surprises Act: verify every 90 days, update within 2 business days	VERIFIED (primary)	"not less frequently than once every 90 days ... verifies and updates"; "within 2 business days". Source: LII, 42 U.S. Code sec. 300gg-115, https://www.law.cornell.edu/uscode/text/42/300gg-115, accessed 2026-10-18
12	provider-data	Medallion median contract ~$45,600 (Vendr)	VERIFIED (third-party)	"median annual contract around $45,600". Source: RFP.wiki, Medallion, https://www.rfp.wiki/vendors/medallion, accessed 2026-10-18
13	premium-audit	NCCI: $41.6B private-carrier WC premium, 2024	VERIFIED (secondary)	"net written premium for private carriers in CY 2024 declined by 3.2% ..., totaling $41.6 billion." Source: Insurance Business, https://www.insurancebusinessmag.com/us/news/workers-comp/workers-comp-premiums-dip-but-combined-ratios-stay-strong--ncci-552668.aspx, accessed 2026-10-18. NCCI site refused connection.
14	premium-audit	Trade press: audit reviews find costly errors	VERIFIED (old article)	Willis: "about one out of five audits that Willis reviews for clients have clerical errors"; "About 10% of Willis clients that go through the audit review process receive a partial premium refund". Source: Business Insurance, https://www.businessinsurance.com/workers-compensation-audit-reviews-can-reveal-costly-errors/, accessed 2026-10-18. Article date not shown; its examples are from 1996-2000.
15	commission	NextCommission $160/month	VERIFIED with correction	"$160 per month for each pack of five seats", or $1,600/year. Source: NextAgency, Pricing, https://www.nextagency.com/pricing, accessed 2026-10-18
16	commission	Commission Tracker $67-119/user/month	NOT FOUND	GetApp returned 403. Source: https://www.getapp.com/sales-software/a/commission-tracker/, accessed 2026-10-18
17	commission	AgencyBloc from $109/user/month	NOT FOUND (vendor quotes only)	Vendor page: "Packages are based on volume of transactions", request pricing. Source: AgencyBloc, Pricing, https://www.agencybloc.com/pricing/, accessed 2026-10-18

#	Venture	Claim on card	Status	Source line
18	commission	~37,000 US independent agencies	VERIFIED (secondary)	"roughly 37,000 independent insurance agencies are operating in the US" (2026 Agency Universe Study, Future One / Big "I", 2026-09-23). Source: Insurance Business, https://www.insurancebusinessmag.com/us/news/cyber/threefourths-saw-revenue-rise-ai-adoption-tripled-since-2024--big-i-study-590958.aspx, accessed 2026-10-18
19	escheatment	ReportMyUP from $299/year	VERIFIED	"$299 /Year" (ReportMyUP) and "$799 /Year" (Plus); Enterprise is custom. Source: ReportMyUP, Pricing, https://www.reportmyup.com/pricing, accessed 2026-10-18
20	chargeback	Chargeflow 25% of recovered chargebacks	VERIFIED	"Automation: 25% of recovered chargebacks". Source: Shopify App Store, Chargeflow, https://apps.shopify.com/chargeflow, accessed 2026-10-18
21	chargeback	Mastercard: $33.79B (2025) to $41.69B (2028)	NOT FOUND	Mastercard page returned 403. Source: https://www.mastercard.com/gb/en/news-and-trends/Insights/2025/2025-global-chargebacks-outlook.html, accessed 2026-10-18
22	ap-recovery	Liberata 25% of recoveries, 12.5% minimum	NOT FOUND	The listing page shows "12.50 GBP a unit"; the percentage would be in the pricing PDF, which was not read. Source: UK Digital Marketplace, https://www.applytosupply.digitalmarketplace.service.gov.uk/g-cloud/services/845673780445211, accessed 2026-10-18
23	cam-audit	Contingency firms take 30-50%; 40% of CAM reconciliations contain errors	VERIFIED (secondary, unnamed source)	"Contingency-only firms charge nothing upfront but take 30-50% of whatever they recover"; "40% of CAM reconciliations contain errors". Source: Gable, https://www.gable.to/blog/post/lease-audit, accessed 2026-10-18
24	subrogation	Vendor fees 15/18/20%	VERIFIED	"cut-rate contingency fees - 15%, 18%, 20%". Source: Matthiesen, Wickert & Lehrer, https://www.mwl-law.com/when-cheaper-feels-better-but-isnt-the-psychology-behind-subrogations-costliest-mistake/, accessed 2026-10-18
25	subrogation	$15-20B missed per year; 15% of claims (card cites MWL)	NOT FOUND in MWL	CLARA (vendor) attributes "$15 billion in annual missed subrogation opportunities" to an NAIC paper, which would not load. Source: CLARA Analytics press release, 2025-05-07, https://claraanalytics.com/news/clara-analytics-unveils-ai-driven-subrogation-detection-for-insurance-claims/, accessed 2026-10-18
26	warranty	Armatus: 7,500+ dealerships, 20,000+ rate increases	NOT FOUND	Not on the cited DigitalDealer page. Source: https://digitaldealer.com/dealer-gm/retail-warranty-reimbursement-leveling-playing-field/, accessed 2026-10-18. Utah Auto Dealer source not checked.
27	warranty	Dealer attorneys report rising factory audit chargebacks	NOT FOUND	Illinois Lawyer Now returned 403. Source: https://www.illinoislawyernow.com/2026/03/chargeback-letter-from-the-factory-illinois-dealers-should-not-treat-it-like-the-final-word/, accessed 2026-10-18. SE-FI and VADA not checked.
28	demurrage	Dry-bulk demurrage $8-10B/yr; 99% of SOFs manual	VERIFIED (vendor claim)	"dry bulk alone accounts for $8-10 billion in demurrage fees annually"; "99% of statements of facts (SOFs) are still processed manually". Source: Marcura, https://marcura.com/blog/demurrage-claims-leakage, accessed 2026-10-18
29	jib-audit	80+ hour audits; 24-month JOA audit window	NOT FOUND	Weaver returned 403. Source: https://weaver.com/resources/considering-joint-interest-audit-answer-these-questions-first/, accessed 2026-10-18

#	Venture	Claim on card	Status	Source line
30	supplier-rebate	Incumbent pricing quote-only; market unsized	Accepted as stated	The card makes no numeric claim to check.

2. Calibrated scores (sorted by new total)

Anchors used for all 13: - **Pricing:** 5 = two or more verified public prices in the buyer segment. 4 = one verified vendor price plus another verified price or range. 3 = a verified industry range or third-party contract figure only. 2 = named paid incumbents, no price. 1 = none. - **Market:** 5 = primary count and dollar pool, still open. 4 = credible industry-body dollar figure. 3 = verified buyer count, or a verified pool that is mostly paid out. 2 = vendor-only or unverified figure. 1 = unknown. - **Demand:** 5 = law with a dated duty plus primary error data. 4 = a verified law/regulation or primary data. 3 = verified independent trade press or a secondary benchmark. 2 = vendor-only or unverified. 1 = none. - **Proof path:** 5 = a measured, confirmable dollar figure within about 30 days. 4 = findings in weeks, cash in 1-3 months. 3 = cash in 3-12 months or hard data access. - **Reuse:** 5 = same pipeline and same buyer as scoup.ai. 4 = same pipeline, new buyer. 3 = new major layer. - **Risk:** 5 = low risk.

Rank	Venture	Old	Price	Market	Demand	Innov	Proof	Reuse	Risk	**New**	scoup.ai relationship / reuse
1	customs-duty-recovery	31	5	3	4	3	4	5	3	**27**	Same buyer (importers) → **add-on module**. Reuses rules engine, evidence packet, per-document hash, tenancy; DRAFT/broker-signs approval
2	commission-reconciliation	27	4	3	2	3	5	4	4	**25**	**Separate business** (insurance agencies) on the same engine. Reuses rules, packet, hashes, tenancy
3	premium-audit-defense	23	2	4	3	4	3	4	3	**23**	**Separate business** (employers via insurance brokers). Reuses recompute rules, packet, hashes, tenancy; broker multi-client tenancy is new
3	provider-data-accuracy	26	3	2	4	3	4	3	4	**23**	**Separate business** (health plans). Reuses packet, hashes, tenancy; NPPES matching layer is new
5	cam-audit	23	3	2	3	3	3	4	4	**22**	Separate business (multi-site tenants). Contract-vs-bill rules, packet, hashes, tenancy
5	jib-audit	22	2	2	2	4	4	4	4	**22**	Separate business (non-operators). Contract-vs-bill rules, packet, hashes
5	escheatment-compliance	23	4	2	3	2	4	4	3	**22**	Separate (holders/CPAs); ledger rules, packet; 50-state rule base new
5	warranty-recovery	25	2	2	3	3	4	4	4	**22**	Separate (auto dealers); claim-dispute shape, packet, hashes
5	chargeback-evidence	27	4	3	3	1	4	3	4	**22**	Separate (e-commerce); packet only, no recompute
10	demurrage-claims	23	2	3	2	3	3	4	4	**21**	Separate (charterers/owners), not scoup.ai container D&D; interval math, packet
10	ap-recovery	25	2	2	2	2	5	4	4	**21**	Same finance buyer type as scoup.ai shippers → could be a module; matching rules, packet
10	subrogation-recovery	24	3	2	3	3	3	4	3	**21**	Separate (P&C carriers); claim/demand shape, packet, hashes

Rank	Venture	Old	Price	Market	Demand	Innov	Proof	Reuse	Risk	New	scoup.ai relationship / reuse
13	supplier-rebate-tracker	23	2	1	2	3	4	4	4	**20**	Separate (distributors); contract-vs-invoice rules, packet

What the platform actually has today (freight-recovery/ARCHITECTURE.md): - **Built:** rules engine, evidence packet with a per-document sha256, and API-key tenant isolation. - **Not built:** a hash-chained audit trail and an approval workflow. The cards say "hash-chained" packet, but today there is only a per-document hash. "Approvals" today means DRAFT output with a human in the loop.

One line of reasoning per score:

customs-duty-recovery (27) - Price 5: Alliance 2%/$10k/$100k (verified) and drawback 15-30% (verified). - Market 3: $166B and 330k importers verified, but $134.7B was already accepted and $122B certified by 2026-09-17; the lasting market (PSC, protest, 1520(d), drawback) is unsized. - Demand 4: the Supreme Court ruling and CAPE are verified, and 4.77M entries failed, but the IEEPA rush is mostly served. - Innov 3: brokers file IEEPA cheaply; the room left is multi-route look-back and the protest/court cases (past-90-day entries, Phase 3). - Proof 4: an ACE export gives findings in weeks; cash for non-IEEPA routes takes months. - Reuse 5: same importer buyer as the scoup.ai importer-first plan. - Risk 3: broker licensing (19 USC 1641), 1592 penalty exposure, policy volatility.

commission-reconciliation (25) - Price 4: NextCommission verified ($160 per 5 seats); the other two prices were not found. - Market 3: 37,000 agencies verified; spend is our own arithmetic. - Demand 2: the 3-6% loss is a vendor-blog claim. - Innov 3: trackers exist; per-line evidence against contracted rates is not clearly sold. - Proof 5: one agency, one statement cycle, the owner decides. - Reuse 4: same engine, new buyer. - Risk 4: moderate PII, no license needed.

premium-audit-defense (23) - Price 2: contingency firms exist; no percentage published. - Market 4: $41.6B NCCI premium verified. - Demand 3: broker data says 1 in 5 audits have clerical errors (verified, but an old article). - Innov 4: services only; no worksheet-recompute software found. - Proof 3: recompute takes days; a carrier credit takes 1-3 months or more. - Reuse 4: same recompute-and-dispute engine, new buyer. - Risk 3: some states regulate fees for these services (Lund Report case, per card).

provider-data-accuracy (23) - Price 3: Vendr median $45.6k (third-party). - Market 2: $2B is a second-hand vendor figure, and CMS puts ongoing compliance cost at ~$253 per entity per year. - Demand 4: the 90-day NSA duty and CMS-4208-F2 are verified, but the date is 2026 not 2027, and 48.74% was not found. - Innov 3: crowded; a provenance ledger is the opening. - Proof 4: NPPES is public, so a measured error rate is possible in days. - Reuse 3: the NPPES matching layer is new. - Risk 4: no PHI.

cam-audit (22) - Price 3: the 30-50% industry band is verified. - Market 2: vendor claim only. - Demand 3: 40% error rate (secondary). - Innov 3: rival software is demo-only. - Proof 3: landlord credits take months. - Reuse 4: new buyer. - Risk 4: commercial data.

jib-audit (22) - Price 2: no public fees. - Market 2: one buyer is evidenced; the segment is unsized. - Demand 2: the Weaver source could not be checked. - Innov 4: 100% rule-based review inside the audit window. - Proof 4: prior audit reports act as labels. - Reuse 4: same engine, new buyer. - Risk 4: no PII.

escheatment-compliance (22) - Price 4: ReportMyUP $299/$799 verified. - Market 2: the $4.49B owner-return figure was not re-verified. - Demand 3: state law is the driver; not quantified. - Innov 2: cheap incumbents. - Proof 4: aged ledger, weeks. - Reuse 4: same engine, new buyer. - Risk 3: PII, rule base liability.

warranty-recovery (22) - Price 2: contingency model, no percentage. - Market 2: the Armatus figures were not found on the cited page. - Demand 3: franchise statutes exist; the attorney articles could not be read. - Innov 3: claim-level gap next to Armatus. - Proof 4: dealer data in weeks. - Reuse 4: same engine, new buyer. - Risk 4: light PII.

chargeback-evidence (22) - Price 4: Chargeflow 25% verified. - Market 3: the Mastercard figure could not be read (403). - Demand 3: same source. - Innov 1: saturated, incumbents and processors already automate submission. - Proof 4: past disputes with outcomes. - Reuse 3: packet only. - Risk 4: PCI can be kept out of scope.

demurrage-claims (21) - Price 2: quote-only incumbents. - Market 3: $8-10B is verified but a vendor figure. - Demand 2: vendor only. - Innov 3: port-call cross-check. - Proof 3: relationship-led data access. - Reuse 4: same engine, new buyer. - Risk 4: commercial data.

ap-recovery (21) - Price 2: Liberata 25% was not found; free tools exist. - Market 2: rate only. - Demand 2: APQC figure not re-verified. - Innov 2: crowded; the README in the folder records a DEFER. - Proof 5: standard exports. - Reuse 4: same engine. - Risk 4: financial data only.

subrogation-recovery (21) - Price 3: 15-20% verified. - Market 2: $15-20B not in the cited source; $15B is a vendor attribution. - Demand 3: vendor entry plus trade press. - Innov 3: execution layer is thinner than detection. - Proof 3: cash in 3-12 months. - Reuse 4: claim shape, new buyer. - Risk 3: PII, practice rules.

supplier-rebate-tracker (20) - Price 2: quote-only incumbents. - Market 1: unknown. - Demand 2: the category exists, nothing more. - Innov 3: buy-side gap. - Proof 4: weeks. - Reuse 4: same engine, new buyer. - Risk 4: no PII.

3. Final top 3

1. customs-duty-recovery - rank it as a scoup.ai importer module, not a separate company

- **For:**
- The best-documented money flow of the 13, with a verified public broker fee and drawback range.
- Same buyer, documents and sales motion as the scoup.ai importer-first plan.
- The CBP declaration shows millions of entries failing on checkable faults (duplicates, missing Chapter 99 lines) and on the 90-day reliquidation limit, which pushes entries into protest and court routes.
- **Against:**
- The IEEPA pool is mostly processed: $122B was certified by 2026-09-17.
- The lasting market (PSC, protest, 1520(d), drawback on 232/301 duties) is unsized.
- Brokers already file for 2%.
- As a standalone company it would be selling into a closing window.
- **First real-data proof within 30 days:** one importer (or its broker) exports 24 months of ACE entry summaries plus its CAPE results (accepted and failed entries). We report (a) dollars on failed or unfiled IEEPA entries with the route still open, and (b) non-IEEPA overpayment candidates by route and deadline. The broker confirms which lines they would file. Cash will not arrive within 30 days.
- **Main risk:** the module is sized on a one-time event. Unless the non-IEEPA look-back finds money at a meaningful rate per importer, it is a lead-generator for scoup.ai, not a revenue line.

2. commission-reconciliation - separate business on the same engine

- **For:**
- Fastest proof path: one agency and one statement cycle, and the owner decides.
- The buyer count is verified (37,000 agencies).
- A verified public price shows agencies pay for commission software.

- Direct engine reuse.
- **Against:**
- The loss rate (3-6%) is only a vendor claim.
- Price points are low ($160 per 5 seats per month).
- Two of the three card prices could not be verified.
- It is a different buyer from scoup.ai, so it needs its own sales motion.
- **First real-data proof within 30 days:** one owner-operated agency gives 6-12 months of carrier statements and its book export. We report dollars of short, missing or double-paid commission with line evidence, and the agency confirms what it will send to carriers. The first carrier adjustment may land in the next statement cycle.
- **Main risk:** agencies may lose much less than 3-6%. Then the product is a cheap tracker in a crowded field.

3. premium-audit-defense - separate business, sold through insurance brokers

Tied at 23 with provider-data-accuracy. It wins on Innovation room (4 vs 3), and provider-data-accuracy lost demand weight on checking: wrong date, missing 48.74% figure, and the CMS estimate that the duty costs ~$253 per year per plan. - **For:** - Large verified premium base ($41.6B). - Broker evidence that errors are common (1 in 5 audits). - Only people-heavy contingency services today. - The findings are arithmetic and need no judgment. - Brokers reach many employers at once. - **Against:** - No public price anywhere. - The error-rate evidence is from an old article. - Carrier credits are slow. - Some states regulate fees for these services. - **First real-data proof within 30 days:** one broker supplies the last 1-3 audit worksheets, payroll by class, declarations and COIs for 5-10 client employers. We recompute and report dollar differences per employer, and the broker marks which they would dispute. - **Main risk:** the share of worksheets with dollar errors big enough to dispute may be small. Contingency pricing may also be restricted in key states, which would leave no business model.

4. What changed versus the batch scores

- **One scale.** Batch 1 scored generously; for example, Reuse 5 was given to any venture with the same engine. Here Reuse 5 needs the same engine and the same buyer as scoup.ai, so only customs gets it. This alone took 1 point off several batch-1 and batch-3 ventures.
- **customs 31 → 27.** Market 5 → 3 and Demand 5 → 4: by September 2026 most of the IEEPA pool was already accepted ($134.7B) or certified ($122B). Proof 5 → 4. It is still first.
- **provider-data-accuracy 26 → 23.**
- The CMS-4208-F2 date is contradicted: the rule applies from 1 January 2026, with 30-day updates and annual attestation.
- 48.74% was not found in the rule.
- The CMS burden estimate (~$253 per entity per year) undercuts the "big spend" story.
- Reuse 4 → 3 because the matching layer is new.
- **warranty 25 → 22 and subrogation 24 → 21.** Their main market figures were not on the cited pages.
- **chargeback 27 → 22.** The Mastercard source could not be read (403), and Reuse was cut to 3. Innovation stays 1.
- **ap-recovery 25 → 21.** The Liberata 25% fee was not on the listing.
- **supplier-rebate 23 → 20.** The market has no figure at all.
- **commission 27 → 25.** AgencyBloc and Commission Tracker prices were not found; the NextCommission price is per 5 seats.
- **premium-audit-defense held at 23 and moved up to third.** Its key figures (NCCI premium, broker error data) checked out.
- **Spread.** Across the 13, the spread is now 20-27. The top three did not change in a way that one point of judgment alone would flip, except the 3rd-place tie, which is explained above.

commission-reconciliation - build plan to a pilot-ready MVP

Goal: run the 30-day proof with one owner-operated agency (6-12 months of carrier statements + book export → dollars of short / missing / double-paid commission with line evidence → agency confirms what it sends to carriers). Inputs: `health.md` (all local suites green; CI blocked by GitHub billing), `gaps.md` (top gaps G1-G6). Principle: smallest path first. **No hosting for the proof**: the analyst runs the CLI offline on an encrypted machine and hands the agency a packet plus spreadsheet. Hosting comes after the proof. Estimates are engineering days for one developer, assuming the partner sample is in hand where noted. Writers work in their own worktree, per hive rules.

Order and dependencies

```
WP0 (owner/legal) ----------------------------------------------+
WP1 (owner: sample request) -> WP2 -> WP3 -> WP4 -> WP5 -> WP6 -> WP8 (dry run) -> pilot
                                    \-> WP7 (privacy, parallel)
CI-0 (owner billing) any time; WP9/WP10 only after the proof
```

WP	Title	Days	Needs owner	Legal	Needs partner sample
CI-0	Unblock GitHub Actions billing; re-run master	0.1	**yes**	no	no
WP0	Pilot terms: confidentiality / data-handling letter, retention, offline processing	0.5 dev + counsel	**yes**	**yes**	no
WP1	Sample request kit + intake checklist	0.5	**yes** (sends it)	no	no
WP2	Source-mapping config (CSV/XLSX headers, carrier + writing-company aliases, policy-number normalisers)	3	no	no	**yes**
WP3	Per-carrier PDF statement parsers for carriers in the sample	2 per carrier (assume 3 = 6)	no	no	**yes**
WP4	Term-level commission ledger (installments, endorsements, window edges, agency-bill exclusion)	4	no	no	partly (lag, bill types)
WP5	Near-duplicate / double-pay detection	1.5	no	no	no
WP6	Confirm-and-send output: decisions file, per-carrier inquiry export, headline dollars table	3	no	no	no
WP7	Privacy: drop insured names at ingest, purge command, run log	1.5	no	review	no
WP8	Labelled dry run on the partner sample + proof metrics	2	yes (review session)	no	**yes**
WP9	(post-proof) Webapp in CI + hosted pilot path	5+	yes	yes	no
WP10	(post-proof) AMS / Ivans download readers	UNKNOWN	yes	maybe	yes

Critical path to pilot-ready: about 22 dev days (WP1 0.5 + WP2 3 + WP3 ~6 + WP4 4 + WP5 1.5 + WP6 3 + WP7 1.5 + WP8 2), plus counsel time for WP0. WP3 scales with carriers in the sample; CSV-only carriers cost ~0.5 day each instead of 2.

CI-0 Unblock CI (owner)

- What: fix the payment / spending limit on the `socrates6969` GitHub account, then re-run run 37336118107. Merge or close Dependabot PRs #1 #2 #5 once green; hold #4 (python 3.14 base image) until tested.
- Acceptance: `gh run list` shows master `ci` green on HEAD 48c8f7a or later.

WP0 Pilot terms (owner + legal)

- What: one-page confidentiality and data-handling letter for a US agency: purpose limited to reconciliation; offline processing on an encrypted machine; no hosted LLM; insured names not required (agency may redact); deletion within N days after the report; findings are DRAFT; the agency decides what goes to carriers; no premium-trust or money movement.
- Files: new `business/pilot-letter-us.md`; `business/dpa-draft.md` stays for EU only.
- **Legal review required:** state insurance data-security duties on the agency as licensee and on us as service provider (state adoption of NAIC Model 668: UNKNOWN), retention term, liability wording for findings, any fee contingent on recovered commission.
- Acceptance: signed letter before any file is received.

WP1 Sample request kit (owner sends)

- What: a 30-minute checklist for the agency: (1) AMS book export for policies in force during the window: policy number, carrier / writing company, LOB, eff/exp, written premium, new/renewal, status, cancel date, bill type (direct/agency), producer code, expected rate if stored; (2) every carrier commission statement for 6-12 months as the carrier provides it; (3) commission schedules or the rate page of each agreement; (4) carrier list with usual pay lag; (5) a contact for questions.
- Files: new `docs/pilot/sample-request.md`, `docs/pilot/intake-checklist.md`.
- Acceptance: partner returns files; the WP2 intake report lists per file: recognised type, rows, period range, carrier, unrecognised columns.

WP2 Source-mapping config (needs sample)

- What: per-partner mapping file: header aliases per source, XLSX sheet selection, carrier and writing-company alias table, policy-number normalisers (prefixes, suffixes, term suffix, leading zeros), bill-type column. Generic code, config per partner.
- Files: `ingest/columns.py` (extra aliases), new `ingest/mapping.py`, `ingest/loader.py` (XLSX via openpyxl: new pinned, hash-locked dependency), `rules/common.py` (`same_carrier`, `norm_id` use the config), `config.py` (`CMR_MAPPING_FILE`), `cli.py` (`--mapping`, `--intake`).
- Tests: renamed header maps via config; XLSX fixture parses identically to its CSV twin; writing-company alias matches; normaliser equates a prefixed/suffixed policy number with the bare one; unknown columns reported; config load deterministic and order-independent.
- Acceptance: on the partner sample, at least 98% of book rows and statement lines parse without warnings, and every failure is listed.

WP3 Per-carrier PDF statement parsers (needs sample)

- What: for each PDF-only carrier, a layout parser (pdfplumber tables / word positions) emitting `CommissionStatement`, with a self-check that lines sum to the printed total.
- Files: new `extraction/carriers/<carrier>.py`, registry in `extraction/service.py`, selection by carrier name or layout fingerprint.

- Tests: golden test per carrier on a redacted real page (only with partner consent; else a synthetic replica of the layout); sum equals printed total; malformed page gives a warning, never a crash; existing hostile-input tests still pass.
- Acceptance: every line on sample statements parsed or flagged; parsed total equals printed total on every statement (or an explicit `COM-STATEMENT-TOTAL` review item).
- Flag: OCR is out of scope; scanned statements are reported as not parsed.

WP4 Term-level commission ledger

- What: new rules `COM-TERM-SHORT` / `COM-TERM-OVER`. Per policy term: expected = (book written premium plus endorsement deltas) x contracted rate; paid = sum of all lines for that term across the window. An unpaid remainder is a confirmed short only when the term is fully billed inside the window and the statements for the due months are present; otherwise pending. Per-line premium mismatch becomes info when lines are installments. Agency-bill terms are excluded from missing. Rate-schedule rules get optional effective dates.
- Files: `rules/engine.py`, new `rules/ledger.py`, `rules/checks.py` (missing / premium-mismatch interplay), `models.py` (bill_type, schedule effective dates, term ledger in the result), `evidence/report.py`.
- Tests: 12 monthly installments summing to the term premium give no finding; 11 of 12 paid with all statements present give a confirmed short of one installment; same with the last month outside the window gives pending; an endorsement raises expected; an agency-bill term is never missing; the exact-sum identity is extended per term (paid + short - over = expected) and property-tested on random installment books.
- Acceptance: on the partner sample, premium-mismatch review noise falls to items a human agrees are real (measured in WP8).

WP5 Near-duplicate / double-pay

- What: detect the same policy + period + type paid twice with different line text or ids, and reversal / repost pairs that net to a double payment. Confirmed only when amounts match exactly and no reversal exists; otherwise review.
- Files: `rules/engine.py` (dedupe stage), `rules/checks.py`.
- Tests: identical rows (existing); same policy and period with different line ids gives double-pay; reversal plus repost gives no finding; partial repost gives review.
- Acceptance: dollars count once (existing invariant holds).

WP6 Confirm-and-send output

- What: (a) headline table of dollars by type and carrier over the window, confirmed vs pending; (b) `decisions.csv` round-trip: one row per finding (finding id, agency decision confirm / reject / needs-info, note); a rerun with the decisions file yields a confirmed-only packet; (c) per-carrier inquiry export (XLSX + CSV): policy, period, expected, paid, difference, rule, file/line reference, packet digest; (d) proof metrics: found, confirmed and rejected dollars by rule.
- Files: `evidence/packet.py`, `evidence/report.py`, new `evidence/export.py`, `cli.py` (`--decisions`, `--export-dir`), `models.py` (stable finding ids).
- Tests: finding ids stable across reruns; decisions change status only, never amounts; exported totals equal packet totals to the cent; content digest stays deterministic; spreadsheet formula-injection guard (cells beginning with a formula character are prefixed).
- Acceptance: the agency opens one file per carrier and marks decisions; we can state "agency confirmed X of Y dollars" from the decisions file.

WP7 Privacy and retention

- What: drop insured-name columns at ingest by default (`CMR_KEEP_INSURED=false`); a purge command for a run directory; a run log (file hashes, operator, time, decisions applied) without document content.
- Files: `ingest/loader.py`, `extraction/stub.py`, `config.py`, `cli.py` (`purge`), `evidence/packet.py`.
- Tests: no insured string appears in the packet, exports or stored raw docs with the flag off; purge removes all run artifacts; the log has no content.
- Legal: retention term comes from WP0.

WP8 Labelled dry run (needs sample + owner session)

- What: run on the partner sample; the analyst hand-checks every confirmed finding and a sample of review items against statements and agreements; the owner walks the agency through findings; decisions are recorded.
- Deliverables: proof report (dollars found by type, confirmed by the agency, precision of confirmed findings, parse coverage, hours spent) and the per-carrier inquiry files the agency will send.
- Acceptance: precision of confirmed findings measured (no accuracy figure is quoted before this); an agency decision recorded for every finding above an agreed dollar threshold.

WP9 / WP10 (after the proof; not on the critical path)

- WP9: webapp job in CI; hosted pilot per `deploy/aws.md` with TLS, rate limiting, malware scan, retention job, pentest; needs the US agreement / DPA (legal).
- WP10: AMS or Ivans download readers once the partner's AMS is known (Applied Epic / AMS360 / HawkSoft / EZLynx export formats are UNKNOWN today).

Blockers and flags

- **Partner sample** blocks WP2, WP3, WP8: no real statement or AMS export layout could be verified.
- **Legal** blocks receiving any data: WP0 letter, state data-security duties, retention, contingent-fee model.
- **Owner** blocks CI: GitHub billing.
- **Research gap:** the WebSearch budget for this session was exhausted; repeat carrier/AMS format research in a fresh session, or rely on the partner's files.

premium-audit-defense - build plan to a broker-pilot-ready build (2026-10-18)

Goal: run the 30-day proof (FINAL.md section 3) on one broker's 5-10 employers x last 1-3 audits, output dollars per employer, collect the broker's "would dispute" marks. Inputs: health.md (code green, CI red on config drift) and gaps.md (top gaps: premium model, input format, state rules, batch/report, licensing+data handling).

Design choices for the pilot (smallest path)

- **Operator-run, local, no hosting.** We (not the broker) run a batch CLI on an encrypted workstation. The existing API, webapp, Docker image and AWS skeleton are not on the critical path.
- **Human transcription first, parsers second.** A validated template is filled from each worksheet; per-carrier PDF parsers are built only for carriers that dominate the real sample.
- **Conservative findings.** Only arithmetic that reconciles every premium component counts as "confirmed"; anything that depends on state rules, class judgment or a missing component is "needs review". The broker decides.
- **Free pilot, broker is the licensed party.** No fee to anyone until counsel signs off (gaps.md gap 5).
- Repo rules apply when building: one writer per worktree, branch off master, tests with every change, no commits to master without the owner.

Work packages (in order)

#	Work package	Files / modules	Acceptance tests	Effort (dev days)	Owner / dependency
WP0	Fix red CI	`.github/workflows/ci.yml` (drop or restore the `developer-portal` pip-audit step and job), `.github/dependabot.yml` (drop `/developer-portal`)	master CI run all green; Dependabot `/developer-portal` job gone	0.25	dev
WP1	Pilot intake kit	new `pilot/intake/checklist.md` (doc request per employer: audit statements 1-3 yrs, declarations + rate pages + mod worksheet, payroll by class/employee export, officer list, subcontractor list + COIs), `pilot/intake/manifest.csv` (employer_id, policy, period, state, carrier, files)	Manifest validator rejects a missing doc type or period; checklist reviewed by the broker	1	dev + founder (broker-facing); **needs broker**
WP9	Licensing gate (parallel from day 1)	`pilot/legal/` notes only	Written counsel view on: NY 2102(b)(3) and the pilot states; free-pilot structure with broker as client; whether a broker may re-bill (Oregon-style "no fees" rules); contingency viability	0 dev; 1-2 weeks calendar	**insurance regulatory counsel (external, cost UNKNOWN)**; LEGAL
WP8	Pilot data handling	`src/premium_audit/pilot/pseudonymize.py` (drop names/SSNs/addresses from payroll registers, keep employee_ref, class, gross, OT premium, severance, officer flag), `pilot/legal/nda-dpa-outline.md`, `pilot/RETENTION.md`, deletion log CSV	Unit tests: SSN/name columns never appear in any output or stored file; deletion script removes raw files and logs hashes; packet still reproducible from pseudonymised input	1.5	dev; NDA/DPA text needs counsel (LEGAL)

#	Work package	Files / modules	Acceptance tests	Effort (dev days)	Owner / dependency
WP3	Structured transcription template + validator	`src/premium_audit/ingest/template.py`; `templates/worksheet_lines.csv`, `premium_components.csv`, `payroll_by_class.csv`, `officers.csv`, `subcontractors.csv`, `declared_rates.csv` with a data dictionary	Validator: row-level errors with file/row/column; line premium cross-check on entry; round-trip a transcribed copy of fixture pa1001 and get the same packet totals; every number keeps a source pointer (page/line)	2	dev; template fields finalised **after first real worksheet**
WP2	Real premium waterfall	`models.py` (premium components, rating periods, multi-state), `rules/defense.py` (recompute: manual premium → ELs/waiver -> mod (by effective date) → schedule rating -> premium discount → expense constant → terrorism/catastrophe → state surcharges/taxes → minimum premium → estimated vs audited → additional/return)	Synthetic realistic worksheets (3+) reconcile to the cent with zero findings; each component wrong by $X yields exactly one finding of $X; unexplained residual over tolerance → "needs review", never confirmed; existing 171 tests stay green	3-4	dev; component formulas need **licensed premium auditor review** (2-4 h, paid)
WP6	Batch run + per-employer report	`src/premium_audit/batch.py` (+ CLI `python -m premium_audit.batch <manifest>`), XLSX/CSV writer (stdlib csv; XLSX via openpyxl only if broker wants it, hash-locked)	Run over a folder of 10 synthetic employers x 3 years: one row per employer-year (billed, recomputed, confirmed $, pending $, top 3 findings, packet hash) + portfolio totals; deterministic output	2	dev
WP7	Broker marks + proof metrics	`src/premium_audit/pilot/marks.py`; report columns `would_dispute` (yes/no/maybe), `reason`, `reviewer_minutes`	Import of a filled report computes: share of employers with confirmed $ > threshold, $ distribution, broker-agreement rate per rule (precision), review time; rejects rows whose packet hash changed	1	dev
WP5	State/bureau rule pack v1 (pilot states only)	`src/premium_audit/rules/params/<state>_<year>.json` (overtime treatment, officer min/max payroll, severance, uninsured-subcontractor charge basis, mod effective-date rule) each with `source` and `as_of`; loader; new rules OFFICER-LIMIT, SPLIT-PERIOD, COI-PERIOD	A parameter with no source loads as UNKNOWN and disables its rule (finding → needs review); tests per state on synthetic cases; reviewer sign-off recorded in the file	3-5	dev + **licensed premium auditor**; **blocked until pilot states are known**; NCCI/WCIRB manuals were unreachable here, so parameters must come from the manuals the reviewer holds
WP4	Carrier PDF parsers (top 1-2 carriers in the sample)	`src/premium_audit/extraction/carriers/<carrier>.py` using pdfplumber tables; output = same model as WP3	Each parser reproduces the human transcription of the real samples exactly; unknown layout falls back to WP3 with a clear message	3-5 (per carrier ~2)	dev; **blocked until real samples arrive**; skip if transcription is fast enough
WP10	(Post-proof, not needed in 30 days) hosted broker tenancy	broker → many employer tenants, persisted review decisions, S3 + KMS, retention jobs, webapp portfolio view, pen test	-	10+	dev + security review

Critical-path effort before data arrives (WP0, WP1, WP8, WP3, WP2, WP6, WP7): about 11-12 dev days. After samples/states arrive: WP5 3-5 days, WP4 0-5 days. Total to pilot-ready: about 14-22 dev days, plus the counsel memo and 2-4 hours (WP2) + 2-4 hours (WP5) of a licensed premium auditor. Effort figures are estimates.

30-day calendar (assuming a broker signs on day 0)

- Days 1-5: WP0, WP1, WP8, WP3 skeleton; WP9 counsel engaged; NDA/DPA with broker; first 2-3 real worksheets in.
- Days 6-12: WP2 against real component lists; WP3 template finalised; WP6.
- Days 10-18: transcribe or parse all 5-30 worksheets; WP5 for the pilot states; reviewer pass on rules.
- Days 18-22: batch run, per-employer report to broker; WP7.
- Days 22-30: broker marks findings; metrics; decision memo (go / no-go on dollars per employer and precision).
- Carrier credits or re-audit outcomes (1-3 months, per the scoring card) fall outside the 30 days; record them later.

Proof acceptance (what "done" means)

- Every supplied worksheet either reconciles or has each dollar difference attributed to a named finding or an explicit "unexplained residual".
- Per-employer report delivered with packet hashes; broker marks returned for every finding.
- Metrics computed: employers with confirmed $ above a threshold agreed with the broker up front, total confirmed $, pending $, broker-agreement rate per rule, reviewer minutes per employer.

Legal / licensing flags

- LEGAL-1: NY Insurance Law 2102(b)(3) makes paid policy review/advice a licensed activity (agent, broker or consultant) with listed exemptions (gaps.md S2). Any fee model touching NY employers needs counsel first.
- LEGAL-2: Other pilot states: licensing for paid audit review and for contingency fees is UNKNOWN; counsel memo required before any fee.
- LEGAL-3: Broker re-billing: "agents ... cannot charge fees" (Oregon, gaps.md S4) style rules may bar brokers from charging employers for our output; structure as broker-paid software, or free, pending counsel.
- LEGAL-4: NDA/DPA with the broker and, through the broker, consent from each employer to share payroll data.
- LEGAL-5: Output stays labelled DRAFT; the broker (licensed) decides and sends any dispute. We never contact a carrier.

Needs a real broker sample (cannot be done on synthetic data)

- Template fields and premium components actually printed by the broker's carriers (WP2, WP3).
- Which carriers dominate (WP4 go/no-go).
- Pilot states (WP5 scope, WP9 scope).
- Payroll export format and officer/subcontractor separation (WP8 pseudonymiser columns).

What blocks building right now

1. **No broker identified** - no samples, no states, no marks. Synthetic-safe work (WP0, WP2 core, WP3 skeleton, WP6, WP7, WP8) can start today.
2. **No licensed premium-audit reviewer** engaged - WP2 sign-off and all of WP5 depend on one.
3. **No counsel** - required before any fee; not required for a free pilot run, but the NDA/DPA needs review.
4. **State rule sources unreachable from this environment** (ncci.com, wcirb.com blocked; web search budget spent): the reviewer must supply the governing manuals.
5. Not a blocker but note: CI is red until WP0 lands, so pilot changes would merge untested.